

MUSTAFA AHSAN

+92 336 4366278 | mustafaahsan2002@gmail.com | Portfolio | LinkedIn | Lahore, Pakistan | Open to remote & relocation

Software engineer building production AI systems: agentic workflows and retrieval over messy enterprise data

EXPERIENCE

AI Engineer (Junior Consultant)

Jul 2025 – Present

Systems Limited — promoted from Associate Consultant, Jul 2026

Lahore

- Own the agent and retrieval layer across enterprise AI deliveries while wider teams handle platform and integration; client-facing from requirements and architecture demos through UAT and handover training.
- Cut inference cost through response caching, prompt caching, and routing lighter tasks to smaller models, benchmarking accuracy against latency across OpenAI, Anthropic, and others to pick the right model per task.
- Agentic Banking Assistant** — *major Kuwaiti retail bank*
- Delivered the bank's AI assistant, now live **bank-wide to all retail customers** and the first fully agentic banking experience in its market.
- Decomposed transfers into four distinct payment flows — by mobile number, to other banks, to other customers in-bank, and between a customer's own accounts — each a separate **LangChain** deep-agent flow rather than one overloaded prompt.
- Gated every step with validation, confirmation, and authorization checks, so no transfer executes on an ambiguous or unverified instruction.
- Validated the flows with a scripted test suite and client UAT before go-live, then instrumented production tracing to catch failures and edge cases in live conversations.
- Paired the assistant with a **LangGraph** retrieval flow that answers product and policy questions end to end in **under 5 seconds** with token-level streaming.

Enterprise Knowledge Assistant — *multi-country telecom group (Zain)*

- Made **10,000+** inconsistently formatted legacy SharePoint documents searchable across operating companies in the Middle East and Africa, delivered through a customized **OpenWebUI** front end with multilingual semantic search.
- Enforced document-level access at query time from each user's Microsoft Entra group membership, and linked related passages across documents in **Neo4j** so users could compare policies and figures between operating companies.
- Extended it with in-chat generation of PDF, Word, and Excel deliverables and rendered charts, plus tool calls that run calculations and filtering over spreadsheet data.
- Cut customer-care lookups to **p95 under 5 seconds** with a dedicated retrieval flow for package and plan questions, replacing manual searches through internal documentation.

Admissions Chatbot — *large Kuwaiti university*

- Shipped a public chatbot in **n8n**, live on the university website, where applicants check their own application status through API-backed flows; a polling job re-ingests source content so degree and deadline answers stay current.

AI Engineer

2024 – 2025

AxonBuild

Remote

- **Thymus Alpha** — shipped a medical learning assistant to **5000+** users: RAG over a large medical corpus with auto-generated MCQs, concept explanations, and diagram retrieval, on FastAPI, Qdrant, and Next.js.
- **Cairya** — designed the retrieval engine and document ingestion pipeline behind a voice conversational-AI platform, plus the Next.js admin application for managing its knowledge base.

AI Intern

Jun 2024 – Aug 2024

Bookme.pk

Lahore

- Built recommendation and support services as Dockerized Flask microservices with Redis caching, plus an OCR pipeline on Google Cloud Vision for automated passport data extraction.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL

AI & LLM Systems: OpenAI, Azure OpenAI, Claude, Gemini, LangGraph, LangChain, OpenWebUI, token streaming, tracing and observability

Search & Retrieval: Qdrant, pgvector, Neo4j, hybrid search, re-ranking, RBAC-aware retrieval, multilingual corpora

Backend & Data: FastAPI, Flask, REST APIs, Redis, n8n, PostgreSQL, MongoDB

Cloud & DevOps: Azure, Microsoft Graph/Entra ID, GCP, Docker, Terraform, CI/CD, Git

EDUCATION

FAST National University

2021 – 2025

Bachelor of Computer Science

Lahore

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional [verify]

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate [verify]